Exploratory data analysis of the armed conflict panel

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2026-10-26

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This report explores `final_data.csv`, the analytical data set built in week 3 and verified against an independently written implementation in `compare_final_data.qmd`. The aim is to understand the panel before fitting anything: how complete it is, how the four mortality outcomes are distributed, how common the conflict exposure is, and whether the unadjusted association runs in the direction the paper reports.

1 The shape of the panel

```
tibble(
  Quantity = c("Rows", "Countries", "Years", "Rows per country",
               "Balanced panel"),
```

```
  Value = c(nrow(dat),
            n_distinct(dat$iso),
            paste(min(dat$year), "to", max(dat$year)),
            paste(range(table(dat$iso)), collapse = " to "),
            n_distinct(table(dat$iso)) == 1)
) |>
  kable(caption = "Panel structure")
```

Table 1: Panel structure

Quantity	Value
Rows	3720
Countries	186
Years	2000 to 2019
Rows per country	20 to 20
Balanced panel	TRUE

Every country contributes exactly 20 rows, so the panel is balanced by construction. That matters for the fixed effects models the paper uses: no country is weighted more heavily simply because it has more years.

```
dat |>
  filter(!is.na(region)) |>
  count(region, name = "Countries") |>
  mutate(Countries = Countries / 20) |>
  arrange(desc(Countries)) |>
  kable(caption = "Countries by region")
```

Table 2: Countries by region

region	Countries
Africa	54
Asia	46
Europe	40
Americas	35
Oceania	11

```
sum(is.na(dat$region))
```

```
[1] 0
```

Every country matched a region, so the region labels can be used to stratify without losing anyone.

2 Missing data

Missingness is the first thing to look at, because a fixed effects model drops a country-year when any variable in it is missing.

```
dat |>
  select(matmor, infmor, neomor, un5mor, armedconflict, armedconflict_lag1,
         earthquake, drought, gdp_1000, oecd, pop_dens, urban, age_dep,
         male_edu, temp, rainfall_1000) |>
  summarise(across(everything(), \(x) sum(is.na(x)))) |>
  pivot_longer(everything(), names_to = "Variable",
               values_to = "Missing country-years") |>
  mutate(Percent = round(100 * `Missing country-years` / nrow(dat), 1)) |>
  arrange(desc(`Missing country-years`)) |>
  kable(caption = "Missing values by variable")
```

Table 3: Missing values by variable

Variable	Missing country-years	Percent
matmor	426	11.5
armedconflict_lag1	186	5.0
gdp_1000	62	1.7
infmor	20	0.5
neomor	20	0.5
un5mor	20	0.5
pop_dens	20	0.5
urban	20	0.5
male_edu	20	0.5
temp	20	0.5
rainfall_1000	20	0.5
armedconflict	0	0.0
earthquake	0	0.0

Variable	Missing country-years	Percent
drought	0	0.0
oecd	0	0.0
age_dep	0	0.0

Three patterns stand out.

The 186 missing values of `armedconflict_lag1` are structural: they are the year 2000 rows, where there is no previous year to lag. Any model using the lagged exposure loses the first year of every country.

Several variables are missing exactly 20 times, which is one country missing across all 20 years rather than scattered gaps.

```
dat |>
  group_by(iso) |>
  summarise(across(c(infmor, neomor, un5mor, pop_dens, urban, male_edu, temp),
                   \(x) sum(is.na(x))), .groups = "drop") |>
  filter(if_any(-iso, \(x) x == 20)) |>
  kable(caption = "Countries missing a variable in every year")
```

Table 4: Countries missing a variable in every year

iso	infmor	neomor	un5mor	pop_dens	urban	male_edu	temp
CIV	0	0	0	20	20	20	20
PRI	20	20	20	0	0	0	0

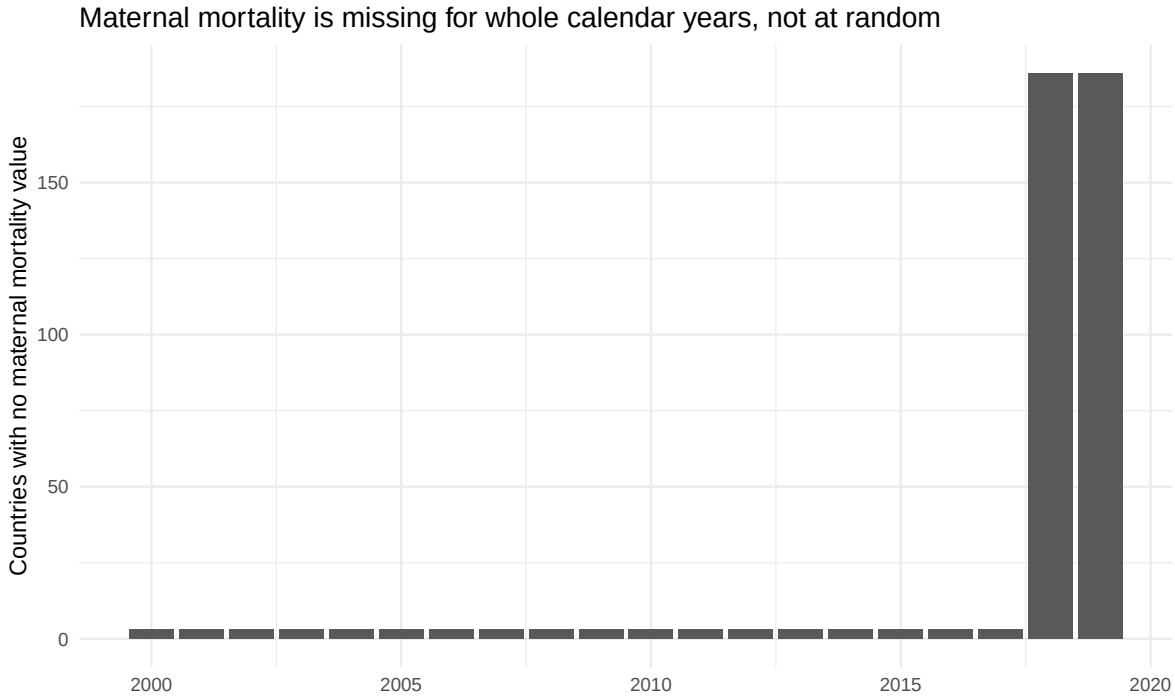
Maternal mortality is the real problem, with 426 missing country-years.

```
dat |>
  group_by(iso) |>
  summarise(missing_matmor = sum(is.na(matmor)), .groups = "drop") |>
  count(missing_matmor, name = "Countries") |>
  rename(`Years of maternal mortality missing` = missing_matmor) |>
  kable(caption = "How maternal mortality missingness is distributed across countries")
```

Table 5: How maternal mortality missingness is distributed across countries

Years of maternal mortality missing	Countries
2	183
20	3

```
dat |>
  group_by(year) |>
  summarise(missing = sum(is.na(matmor)), .groups = "drop") |>
  ggplot(aes(x = year, y = missing)) +
  geom_col(fill = "grey35") +
  labs(title = "Maternal mortality is missing for whole calendar years, not at random",
       x = NULL, y = "Countries with no maternal mortality value")
```



The pattern is not scattered. Almost every country is missing exactly two years, and three countries are missing all twenty. The two missing years are the same two calendar years for everyone, because the World Bank maternal mortality series ends before the child mortality series does.

```
dat |>
  group_by(year) |>
  summarise(`Countries with a value` = sum(!is.na(matmor)), .groups = "drop") |>
  filter(year >= 2015) |>
  kable(caption = "Maternal mortality coverage in the last five years")
```

Table 6: Maternal mortality coverage in the last five years

year	Countries with a value
2015	183
2016	183
2017	183
2018	0
2019	0

So a complete case analysis of maternal mortality is an analysis of 18 years and 183 countries, not 20 and 186. The loss is systematic rather than random, but it is also a clean truncation of the window rather than a pattern that depends on conflict status.

```
dat |>
  group_by(iso) |>
  summarise(any_matmor = any(!is.na(matmor)), .groups = "drop") |>
  count(any_matmor, name = "Countries") |>
  rename(`Has at least one maternal mortality value` = any_matmor) |>
  kable(caption = "Countries usable in a maternal mortality model")
```

Table 7: Countries usable in a maternal mortality model

Has at least one maternal mortality value	Countries
FALSE	3
TRUE	183

3 The four mortality outcomes

```
dat |>
  select(all_of(names(outcomes))) |>
```

```
  pivot_longer(everything(), names_to = "var", values_to = "value") |>
  filter(!is.na(value)) |>
  group_by(Outcome = recode(var, !!!outcomes)) |>
  summarise(N = n(),
            Min = round(min(value), 1),
            Q1 = round(quantile(value, 0.25), 1),
            Median = round(median(value), 1),
            Mean = round(mean(value), 1),
            Q3 = round(quantile(value, 0.75), 1),
            Max = round(max(value), 1),
            .groups = "drop") |>
  kable(caption = "Distribution of each mortality outcome")
```

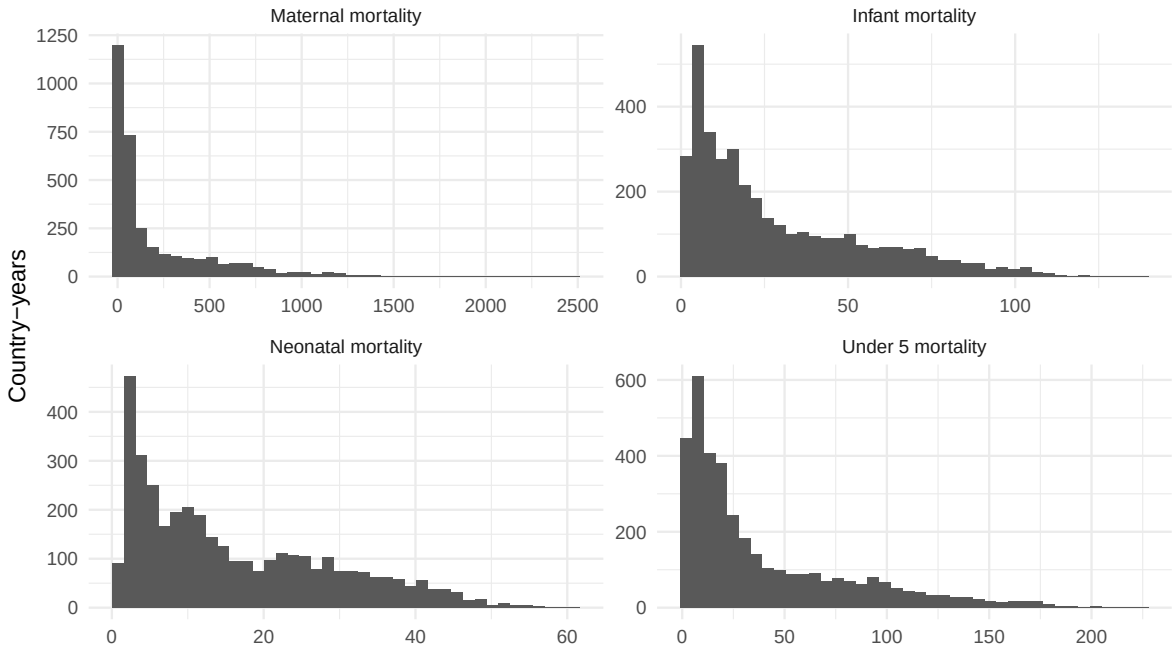
Table 8: Distribution of each mortality outcome

Outcome	N	Min	Q1	Median	Mean	Q3	Max
Infant mortality	3700	1.6	7.6	18.9	28.9	44.5	138.1
Maternal mortality	3294	2.0	17.0	66.0	210.6	299.8	2480.0
Neonatal mortality	3700	0.8	4.9	12.1	16.2	25.3	60.9
Under 5 mortality	3700	2.0	9.0	22.2	40.5	61.3	224.9

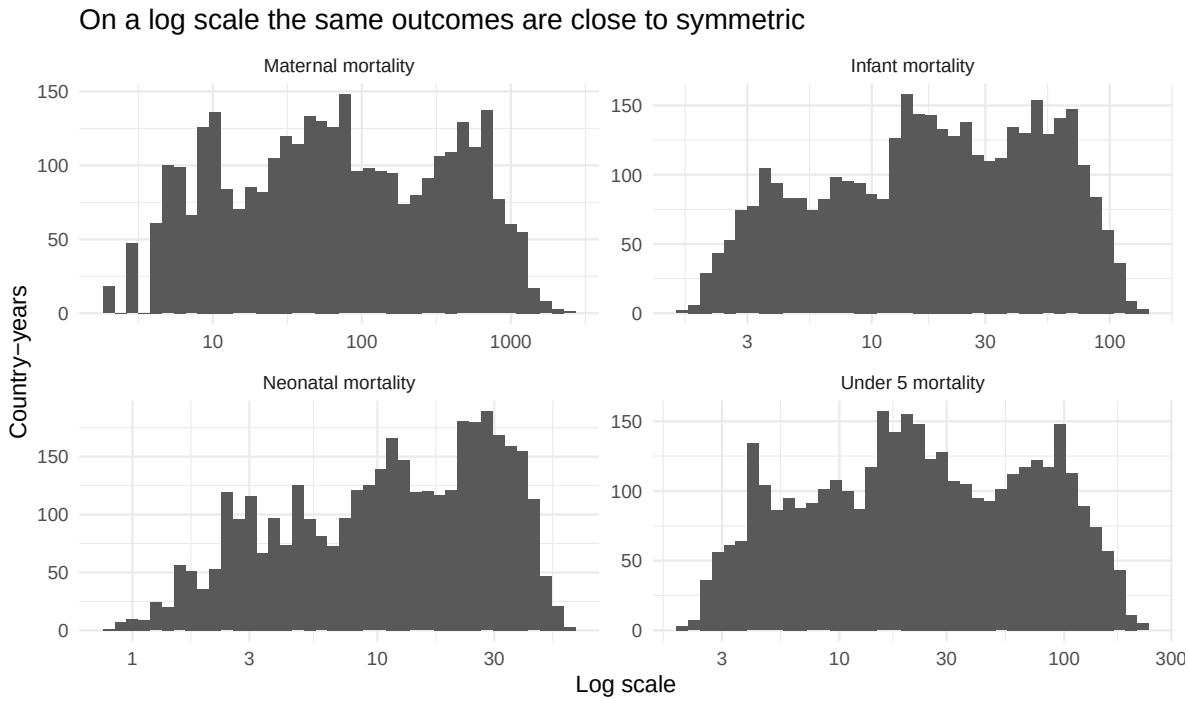
Every outcome has a mean well above its median, which is the signature of a long right tail. Maternal mortality is the most extreme: the median is 66 deaths per 100,000 live births and the maximum is 2480.

```
dat |>
  select(all_of(names(outcomes))) |>
  pivot_longer(everything(), names_to = "var", values_to = "value") |>
  filter(!is.na(value)) |>
  mutate(Outcome = factor(recode(var, !!!outcomes), levels = outcomes)) |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 40, fill = "grey35") +
  facet_wrap(~Outcome, scales = "free") +
  labs(title = "All four outcomes are strongly right skewed",
       x = NULL, y = "Country-years")
```

All four outcomes are strongly right skewed




```
dat |>
  select(all_of(names(outcomes))) |>
  pivot_longer(everything(), names_to = "var", values_to = "value") |>
  filter(!is.na(value), value > 0) |>
  mutate(Outcome = factor(recode(var, !!!outcomes), levels = outcomes)) |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 40, fill = "grey35") +
  scale_x_log10() +
  facet_wrap(~Outcome, scales = "free") +
  labs(title = "On a log scale the same outcomes are close to symmetric",
       x = "Log scale", y = "Country-years")
```



The log scale matters for how the models are specified. On the raw scale a handful of very high mortality country-years would dominate the fit; on the log scale the outcomes are roughly symmetric, which is why the paper models them with a log transformation.

```
dat |>
  select(all_of(names(outcomes))) |>
  cor(use = "pairwise.complete.obs") |>
  round(3) |>
  as.data.frame() |>
  rownames_to_column("Outcome") |>
  kable(caption = "Correlation between the four outcomes")
```

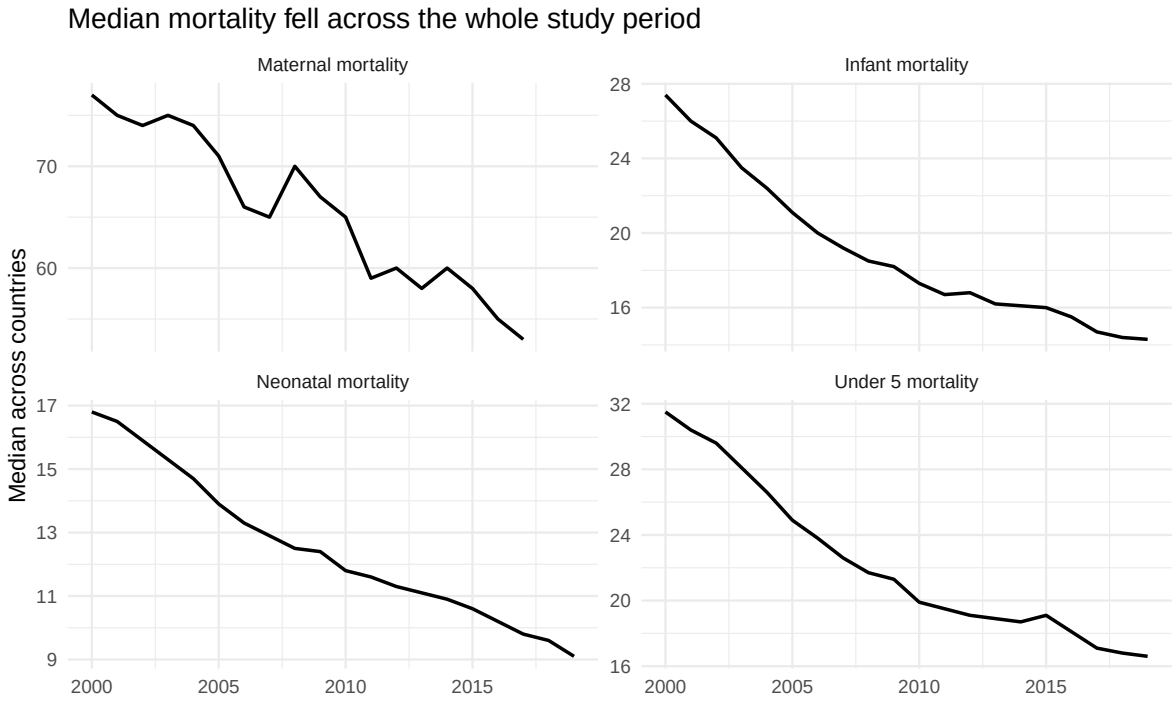
Table 9: Correlation between the four outcomes

Outcome	matmor	infmor	neomor	un5mor
matmor	1.000	0.879	0.835	0.899
infmor	0.879	1.000	0.959	0.986
neomor	0.835	0.959	1.000	0.927
un5mor	0.899	0.986	0.927	1.000

The three child mortality measures are almost the same variable measured three ways, with

correlations above 0.95. Maternal mortality is strongly but less tightly related to them. Treating the four as four independent findings would overstate how much evidence there is.

```
dat |>
  select(all_of(names(outcomes)), year) |>
  pivot_longer(-year, names_to = "var", values_to = "value") |>
  filter(!is.na(value)) |>
  mutate(Outcome = factor(recode(var, !!!outcomes), levels = outcomes)) |>
  group_by(Outcome, year) |>
  summarise(median = median(value), .groups = "drop") |>
  ggplot(aes(x = year, y = median)) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~Outcome, scales = "free_y") +
  labs(title = "Median mortality fell across the whole study period",
       x = NULL, y = "Median across countries")
```



Every outcome declines steadily from 2000 to 2019. A model that ignores calendar time would attribute part of this global decline to whatever else changed over the same period, which is why year effects belong in the model.

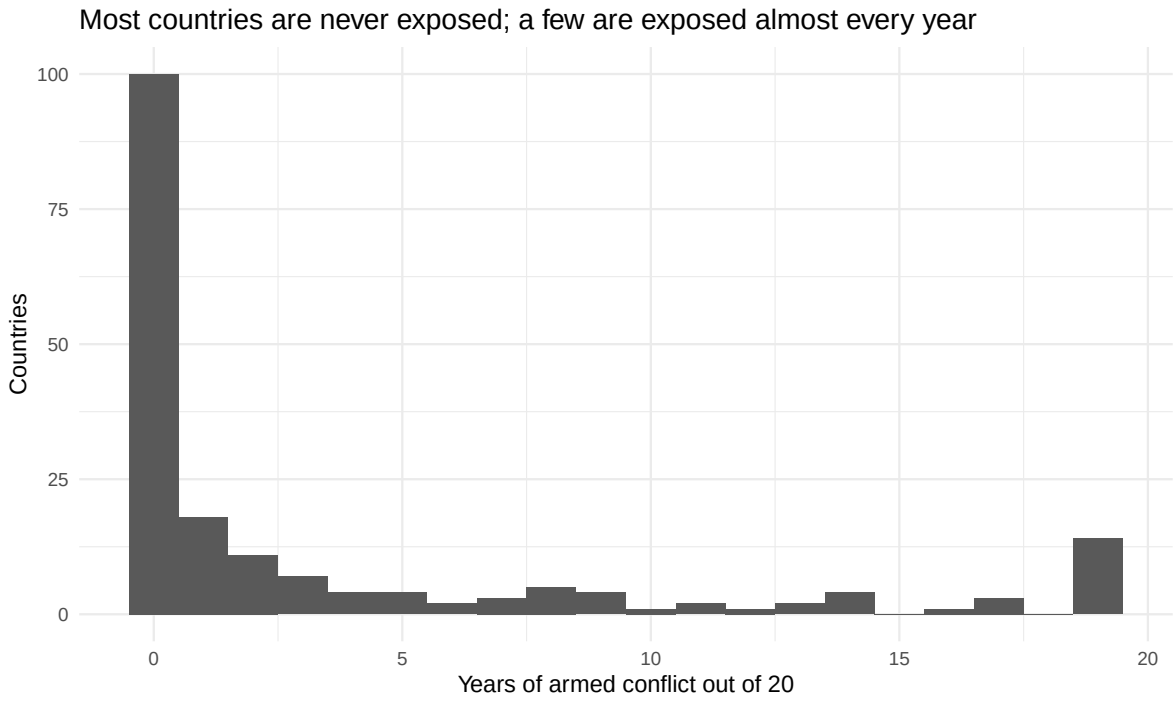
4 The conflict exposure

```
tibble(
  Quantity = c("Exposed country-years",
               "Percent of all country-years",
               "Countries ever exposed",
               "Countries never exposed",
               "Countries exposed in all 20 years"),
  Value = c(sum(dat$armedconflict),
            paste0(round(100 * mean(dat$armedconflict), 1), "%"),
            n_distinct(dat$iso[dat$armedconflict == 1]),
            n_distinct(dat$iso) - n_distinct(dat$iso[dat$armedconflict == 1]),
            sum(tapply(dat$armedconflict, dat$iso, sum) == 20))
) |>
  kable(caption = "How common the exposure is")
```

Table 10: How common the exposure is

Quantity	Value
Exposed country-years	665
Percent of all country-years	17.9%
Countries ever exposed	86
Countries never exposed	100
Countries exposed in all 20 years	0

```
dat |>
  group_by(iso) |>
  summarise(years = sum(armedconflict), .groups = "drop") |>
  ggplot(aes(x = years)) +
  geom_histogram(binwidth = 1, fill = "grey35") +
  labs(title = "Most countries are never exposed; a few are exposed almost every year",
       x = "Years of armed conflict out of 20", y = "Countries")
```



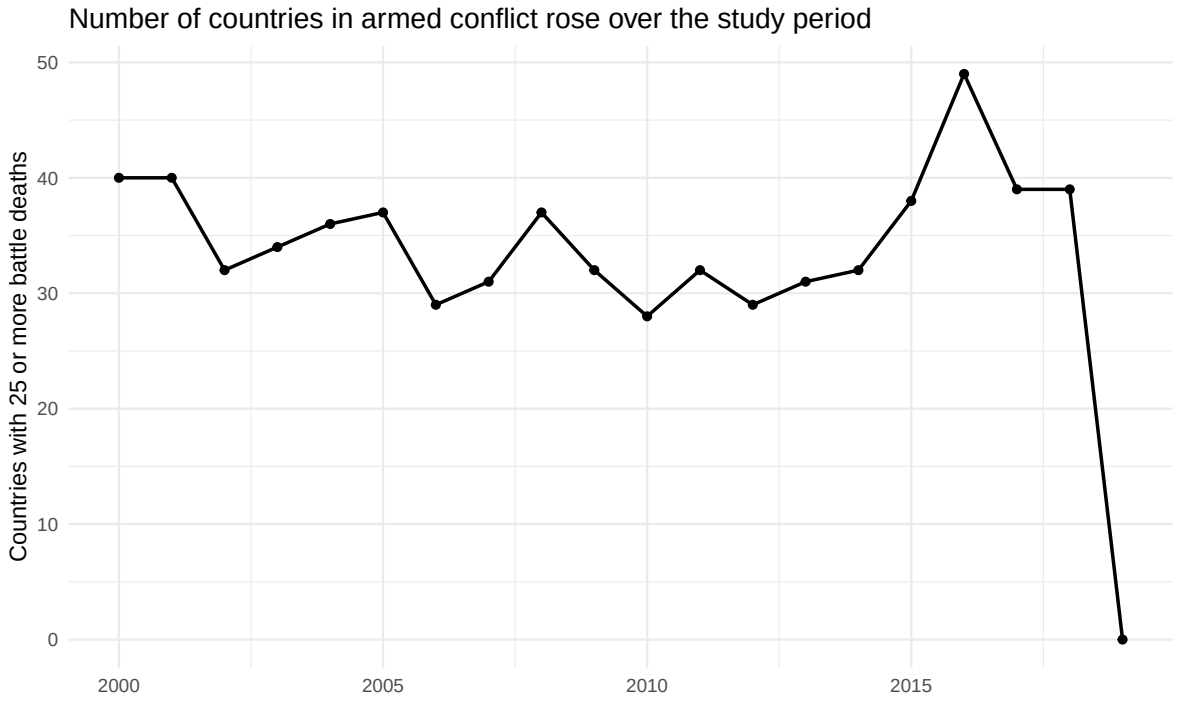
This distribution is the central limitation of the design. Half the countries contribute no exposure variation at all, and a country fixed effects model identifies the effect only from countries that change status during the window.

```
dat |>
  group_by(iso) |>
  summarise(years = sum(armedconflict), .groups = "drop") |>
  summarise(`Never exposed` = sum(years == 0),
            `Changes status at least once` = sum(years > 0 & years < 20),
            `Always exposed` = sum(years == 20)) |>
  pivot_longer(everything(), names_to = "Group", values_to = "Countries") |>
  kable(caption = "Which countries can contribute to a within country comparison")
```

Table 11: Which countries can contribute to a within country comparison

Group	Countries
Never exposed	100
Changes status at least once	86
Always exposed	0

```r
dat |>
  group_by(year) |>
  summarise(exposed = sum(armedconflict), .groups = "drop") |>
  ggplot(aes(x = year, y = exposed)) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 1.4) +
  labs(title = "Number of countries in armed conflict rose over the study period",
       x = NULL, y = "Countries with 25 or more battle deaths")
```




```r
dat |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(`Country-years` = n(),
            Exposed = sum(armedconflict),
            Percent = round(100 * mean(armedconflict), 1),
            .groups = "drop") |>
  arrange(desc(Percent)) |>
  kable(caption = "Exposure by region")
```

Table 12: Exposure by region

region	Country-years	Exposed	Percent
Africa	1080	302	28.0
Asia	920	250	27.2
Americas	700	81	11.6
Europe	800	30	3.8
Oceania	220	2	0.9

Exposure is concentrated in Africa and Asia, which are also the regions with the highest mortality. The two are confounded at the region level, so a comparison that pools countries would mostly be comparing regions rather than measuring conflict.

```
dat |>
  group_by(iso) |>
  summarise(transitions = sum(diff(armedconflict) != 0), .groups = "drop") |>
  count(transitions, name = "Countries") |>
  rename(`Changes in conflict status` = transitions) |>
  kable(caption = "How often countries switch conflict status")
```

Table 13: How often countries switch conflict status

Changes in conflict status	Countries
0	100
1	19
2	25
3	11
4	17
5	3
6	4
7	5
9	2

5 Exposure against the outcomes

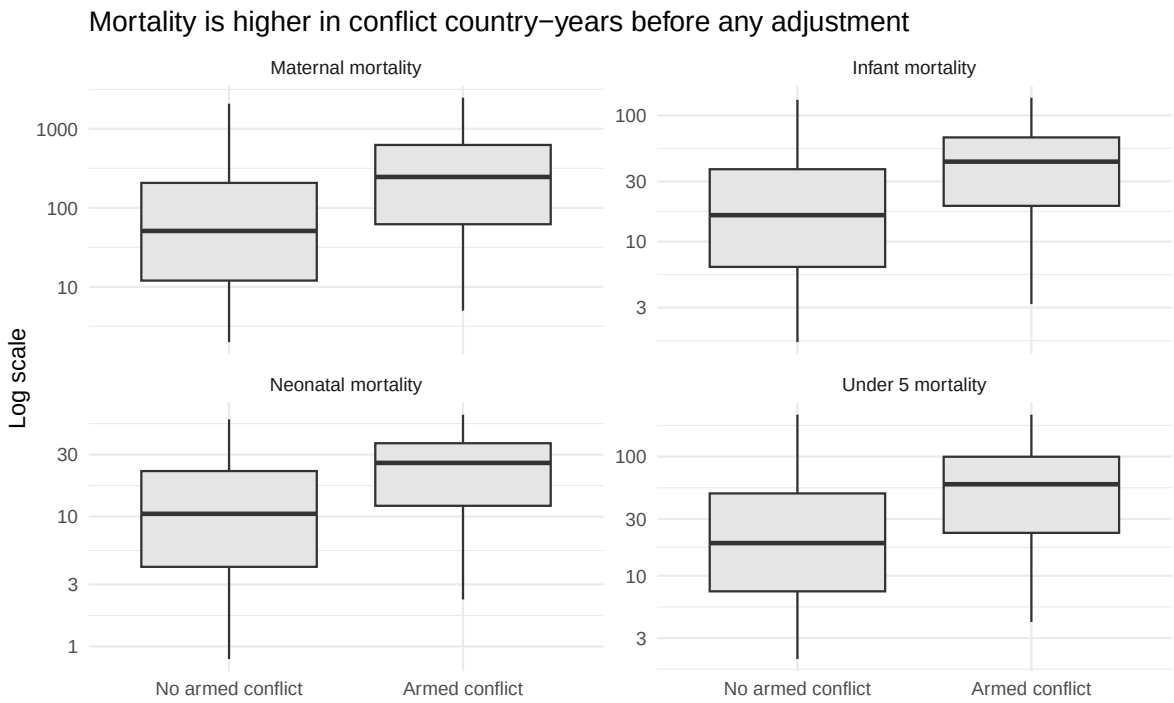
```
dat |>
  select(all_of(names(outcomes)), exposure) |>
  pivot_longer(-exposure, names_to = "var", values_to = "value") |>
```

```
  filter(!is.na(value)) |>
  mutate(Outcome = factor(recode(var, !!!outcomes), levels = outcomes)) |>
  group_by(Outcome, exposure) |>
  summarise(N = n(), Median = round(median(value), 1),
            Mean = round(mean(value), 1), .groups = "drop") |>
  kable(caption = "Each outcome by conflict status, unadjusted")
```

Table 14: Each outcome by conflict status, unadjusted

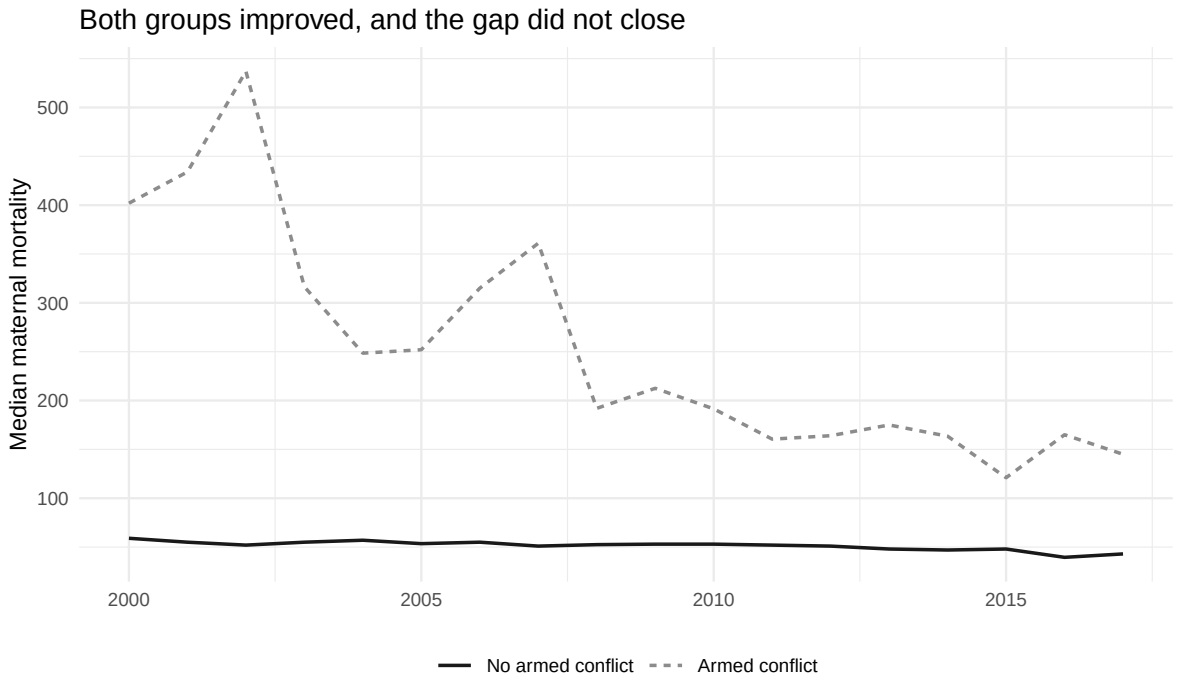
Outcome	exposure	N	Median	Mean
Maternal mortality	No armed conflict	2668	51.0	168.8
Maternal mortality	Armed conflict	626	246.0	388.8
Infant mortality	No armed conflict	3035	16.2	25.1
Infant mortality	Armed conflict	665	43.0	46.2
Neonatal mortality	No armed conflict	3035	10.5	14.2
Neonatal mortality	Armed conflict	665	25.9	25.4
Under 5 mortality	No armed conflict	3035	18.8	34.6
Under 5 mortality	Armed conflict	665	58.6	67.2

```
dat |>
  select(all_of(names(outcomes)), exposure) |>
  pivot_longer(-exposure, names_to = "var", values_to = "value") |>
  filter(!is.na(value), value > 0) |>
  mutate(Outcome = factor(recode(var, !!!outcomes), levels = outcomes)) |>
  ggplot(aes(x = exposure, y = value)) +
  geom_boxplot(outlier.size = 0.5, fill = "grey90") +
  scale_y_log10() +
  facet_wrap(~Outcome, scales = "free_y") +
  labs(title = "Mortality is higher in conflict country-years before any adjustment",
       x = NULL, y = "Log scale")
```



The unadjusted gap runs in the direction the paper reports, and it is large: median maternal mortality is several times higher in exposed country-years. But this comparison is between countries, not within them, and the regional table above shows how much of it could be geography rather than conflict.

```
dat |>
  filter(!is.na(matmor)) |>
  group_by(year, exposure) |>
  summarise(median = median(matmor), .groups = "drop") |>
  ggplot(aes(x = year, y = median, colour = exposure, linetype = exposure)) +
  geom_line(linewidth = 0.7) +
  scale_colour_grey(start = 0.1, end = 0.55) +
  labs(title = "Both groups improved, and the gap did not close",
       x = NULL, y = "Median maternal mortality", colour = NULL, linetype = NULL) +
  theme(legend.position = "bottom")
```




```
switchers <- dat |>
  group_by(iso) |>
  filter(sum(armedconflict) > 0, sum(armedconflict) < 20) |>
  ungroup()

switchers |>
  filter(!is.na(un5mor)) |>
  group_by(exposure) |>
  summarise(Countries = n_distinct(iso),
            `Country-years` = n(),
            `Median under 5 mortality` = round(median(un5mor), 1),
            .groups = "drop") |>
  kable(caption = "Under 5 mortality within the countries that change status")
```

Table 15: Under 5 mortality within the countries that change status

exposure	Countries	Country-years	Median under 5 mortality
No armed conflict	86	1055	39.2
Armed conflict	86	665	58.6

Restricting to the countries that actually change status is the closest this EDA gets to the

comparison a fixed effects model makes, and the difference is far smaller than the pooled one. That is the expected direction: much of the raw gap is between country differences that the fixed effects absorb.

6 The covariates

```
dat |>
  select(gdp_1000, pop_dens, urban, age_dep, male_edu, temp, rainfall_1000) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "value") |>
  filter(!is.na(value)) |>
  group_by(Variable) |>
  summarise(N = n(),
            Median = round(median(value), 2),
            Mean = round(mean(value), 2),
            Min = round(min(value), 2),
            Max = round(max(value), 2),
            .groups = "drop") |>
  kable(caption = "Continuous covariates")
```

Table 16: Continuous covariates

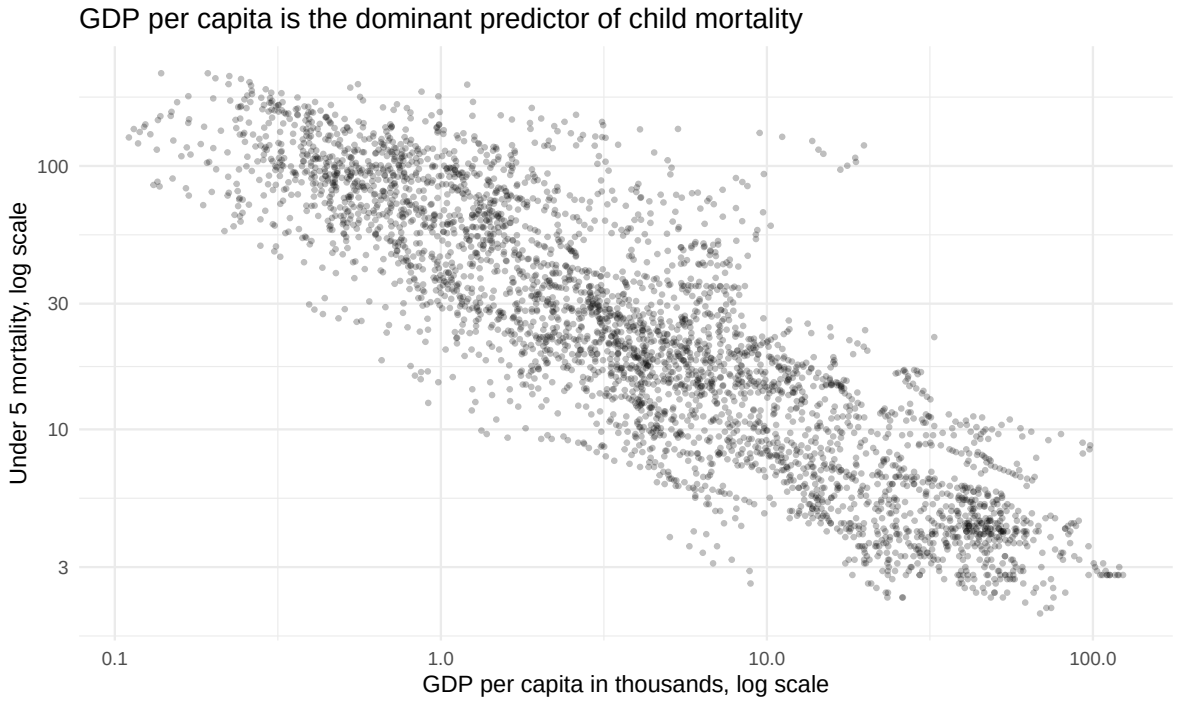
Variable	N	Median	Mean	Min	Max
age_dep	3720	55.51	61.94	16.17	111.48
gdp_1000	3658	4.07	11.49	0.11	123.68
male_edu	3700	8.37	8.26	1.07	14.44
pop_dens	3700	27.52	30.57	0.00	99.86
rainfall_1000	3700	1.01	1.20	0.02	4.71
temp	3700	21.96	19.63	-2.40	29.68
urban	3700	30.25	30.69	0.10	93.41

```
dat |>
  summarise(`OECD country-years` = sum(oecd),
            `Earthquake country-years` = sum(earthquake),
            `Drought country-years` = sum(drought)) |>
  pivot_longer(everything(), names_to = "Indicator", values_to = "Country-years") |>
  mutate(Percent = round(100 * `Country-years` / nrow(dat), 1)) |>
  kable(caption = "Binary covariates")
```

Table 17: Binary covariates

Indicator	Country-years	Percent
OECD country-years	636	17.1
Earthquake country-years	310	8.3
Drought country-years	325	8.7

```
dat |>
  filter(!is.na(gdp_1000), !is.na(un5mor)) |>
  ggplot(aes(x = gdp_1000, y = un5mor)) +
  geom_point(alpha = 0.25, size = 0.7) +
  scale_x_log10() +
  scale_y_log10() +
  labs(title = "GDP per capita is the dominant predictor of child mortality",
       x = "GDP per capita in thousands, log scale",
       y = "Under 5 mortality, log scale")
```




```
dat |>
  select(matmor, infmor, neomor, un5mor, gdp_1000, urban, age_dep, male_edu,
         pop_dens, temp, rainfall_1000) |>
```

```r
  cor(use = "pairwise.complete.obs") |>
  round(2) |>
  as.data.frame() |>
  select(matmor, infmor, neomor, un5mor) |>
  rownames_to_column("Variable") |>
  kable(caption = "Correlation of each covariate with the four outcomes")
```

Table 18: Correlation of each covariate with the four outcomes

Variable	matmor	infmor	neomor	un5mor
matmor	1.00	0.88	0.84	0.90
infmor	0.88	1.00	0.96	0.99
neomor	0.84	0.96	1.00	0.93
un5mor	0.90	0.99	0.93	1.00
gdp_1000	-0.38	-0.50	-0.54	-0.46
urban	-0.32	-0.36	-0.37	-0.34
age_dep	0.77	0.81	0.79	0.82
male_edu	-0.68	-0.74	-0.77	-0.73
pop_dens	-0.32	-0.35	-0.35	-0.34
temp	0.41	0.47	0.48	0.46
rainfall_1000	0.05	0.07	0.05	0.04

GDP per capita, male education and age dependency are all strongly related to the outcomes, and they are also related to each other, so the adjusted estimates will be sensitive to which of them the model includes.

7 What this means for the analysis

The panel is balanced and complete enough to work with, but four things from this EDA should shape the modelling.

Maternal mortality is missing for 426 country-years, and the pattern is structural: the series ends two years before the child mortality series, and three countries have no values at all. A maternal mortality model is fitted on 18 years and 183 countries while the child mortality models get 20 and 186, so the two sets of estimates are not drawn from the same sample.

All four outcomes are strongly right skewed and close to symmetric on the log scale, so they should enter the model logged.

All four outcomes fell steadily from 2000 to 2019, so calendar time has to be in the model or the secular decline will contaminate the exposure estimate.

Only half the countries ever experience armed conflict, and the countries that change status during the window are the only ones that contribute to a within country comparison. The pooled gap between exposed and unexposed country-years is much larger than the gap among those switchers, which is a warning that the unadjusted contrast is mostly between country variation. The fixed effects specification the paper uses is the right answer to that, and it is also why the published effect sizes are far smaller than the raw comparison suggests.